

AI in Marine Science: Where We Are & How We Catch Up

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Outline

- Intro (Fathom Science, myself)
- Landmark Publications in Marine Science
- Marine Science Difficulties
- Compute Difficulties
- A look ahead
- Conclusion



Ruoying (Roy) He, Ph.D. of Oceanography
Founder, President
Fathom Science Inc. 2024 – present



Michael Gray
M.S. Oceanography
Head of Engineering



Taylor Shropshire
Ph.D. Oceanography
Head of Products



Haibo Zong
Ph.D. Oceanography
Head of Modeling



Relocatable, On-demand, AI-Augmented Digital Twin Framework

Integrating physics, AI, and real-time data into decision-ready intelligence



Data
Sources

Earth System
Modeling

AI &
Assimilation

Forecast &
Insight

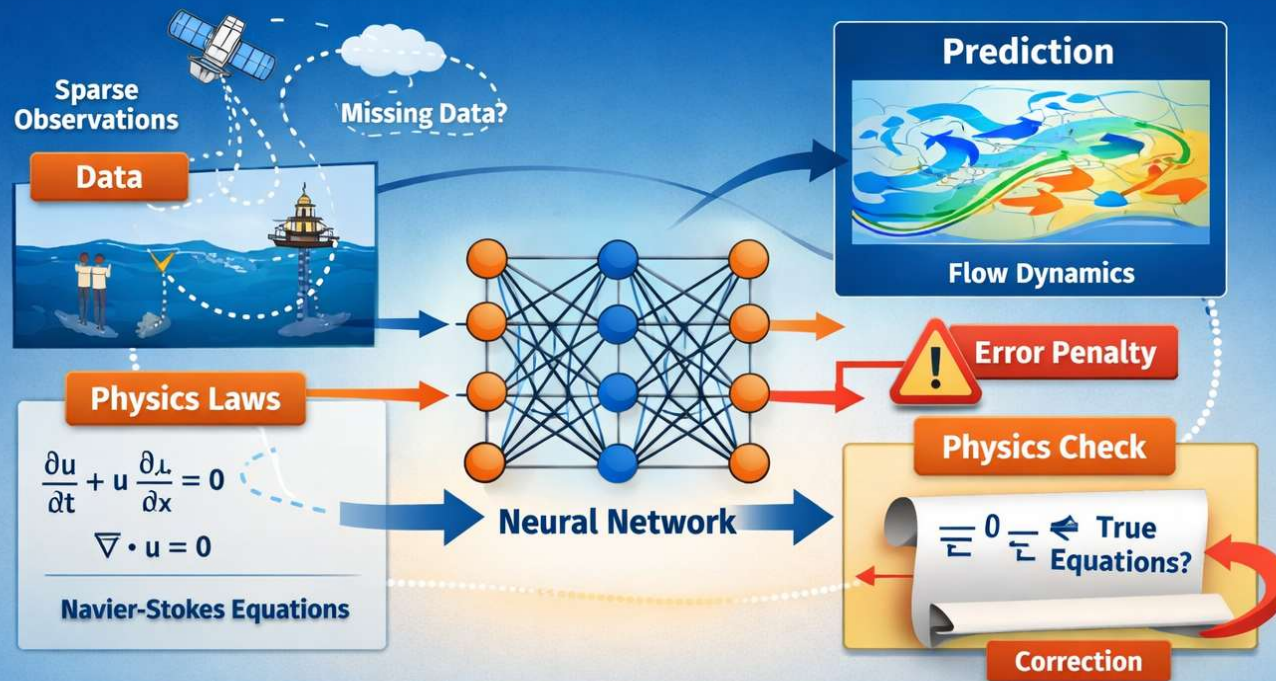
1,000× finer • 500× faster • more accurate

Scientific Foundation



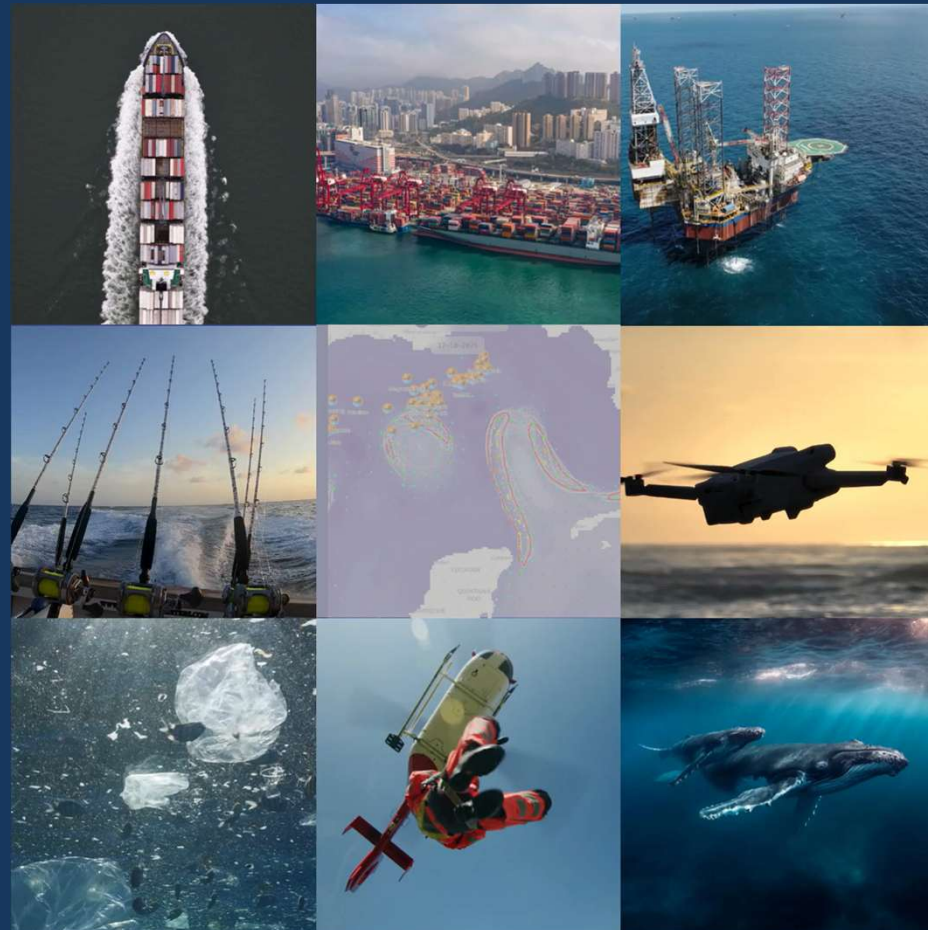
Fathom
Science

Physics-Informed **Neural Network** (PINN)



Trained Model: Predictions Aligned with Physics

A unified digital twin architecture supporting diverse maritime missions



The Future of Predictive Maritime Intelligence



Converging physics, AI, and operations into decision-ready intelligence.

My Background

- BS Physics, Marine Science - NCSU 2021
 - Undergraduate Research w/ Dr. He: analyzing change in ocean heat content following Nor'easters
- MS Marine, Earth, and Atmospheric Sciences – NCSU 2024
 - Numerical modeling, mostly weather
 - Thesis w/ Dr. He
- AI Scientist – Fathom Science 2024



Long-term prediction of the Gulf Stream meander using OceanNet: a principled neural-operator-based digital twin

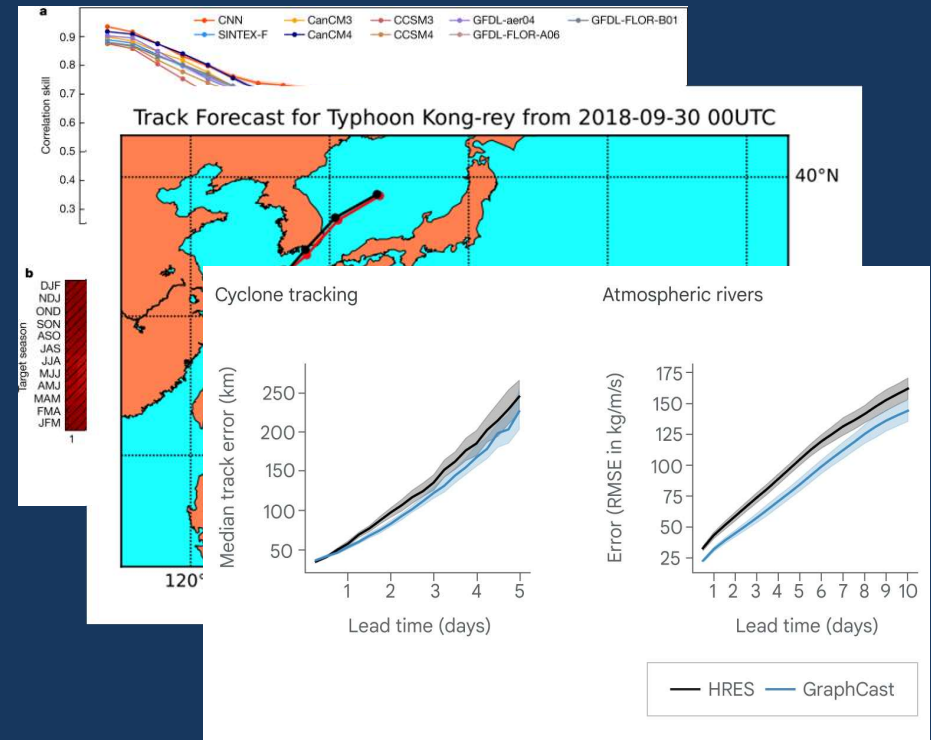
Michael Gray¹, Ashesh Chattopadhyay², Tianning Wu¹, Anna Lowe¹, and Ruoying He¹

¹Marine, Earth, and Atmospheric Sciences, North Carolina State University, Raleigh, NC 27695, USA

²Applied Mathematics, University of California, Santa Cruz, Santa Cruz, CA 95060, USA

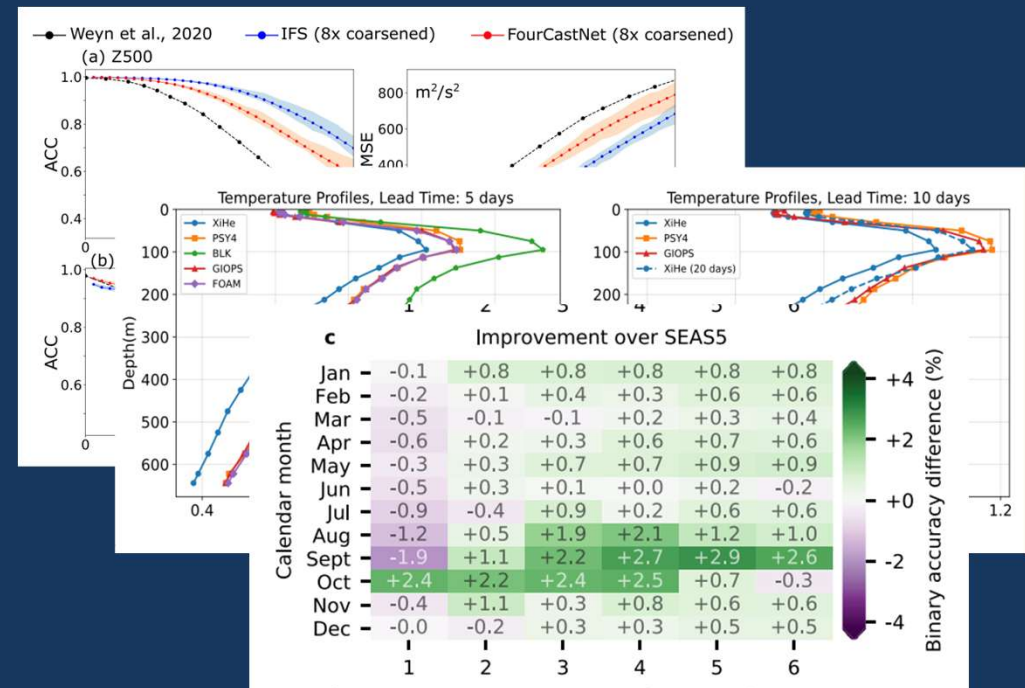
Some Recent Landmarks

- ENSO prediction - CNN + Transfer Learning (Ham et al., 2019)
 - 17-month prediction of ENSO indices, overcoming the spring predictability barrier
- Pangu-Weather – Transformer (Huawei, Bi et al., 2023)
 - First global weather + ocean surface forecast model to surpass numerical models in short- and medium-term forecasts
- GraphCast – GNN (DeepMind/Google, Lam et al., 2023)
 - Scale-invariant medium-range global weather and ocean surface forecasts
 - Open source

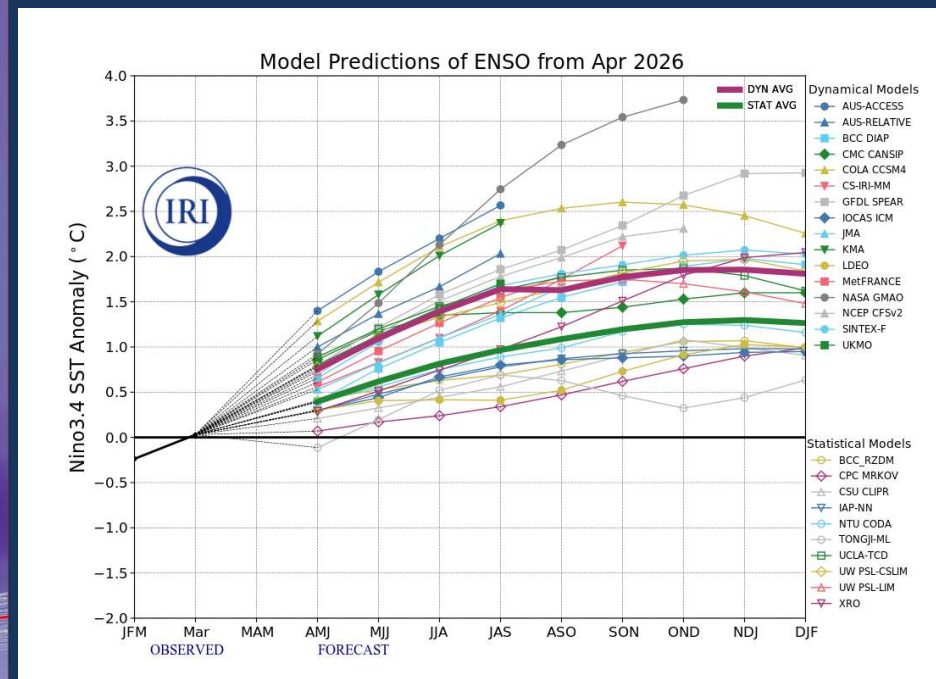


Continued

- FourCastNet – AFNO (NVIDIA, Kurth et. Al, 2023)
 - First of the global forecast models
- XiHe – Transformer (Wang et al., 2024)
 - Global ocean, eddy-resolving forecast
 - Forecast lead surpasses 10d
- FengWu – Multimodel learning (Shanghai AI Lab, Chen et al., 2023)
 - Global weather
 - Forecast lead surpasses 10d
- IceNet – Unet ensemble (Andersson et al., 2021)
 - Monthly sea ice concentrations
 - First model with uncertainty awareness

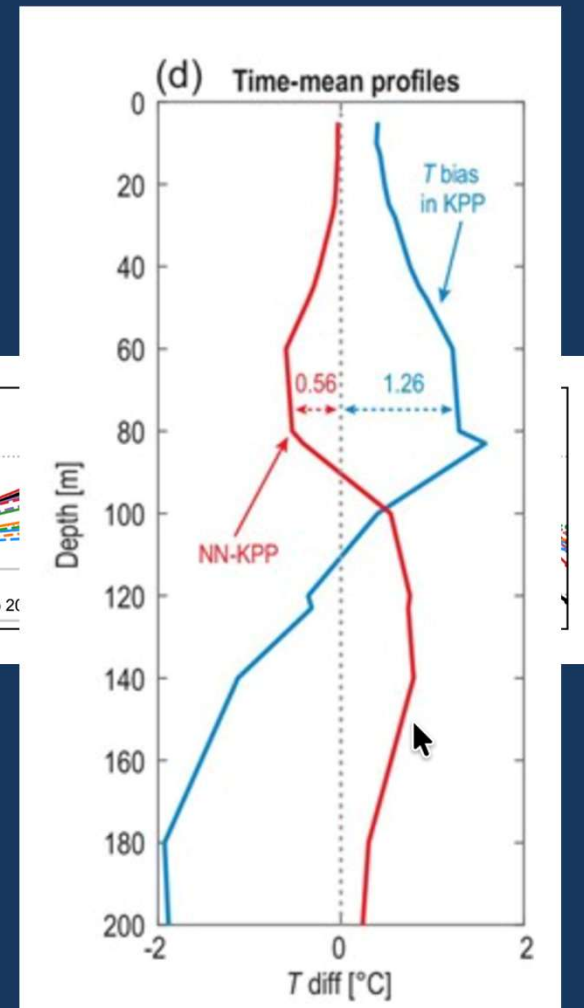
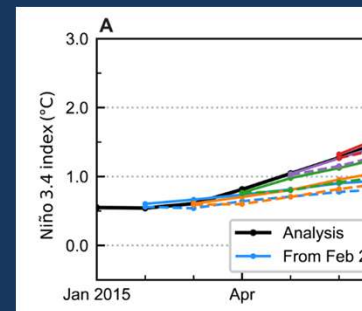


Aside: Ensemble Modeling



Continued

- 3D-Geoformer – Data Driven Transformer w/ spatiotemporal self-attention (Zhou & Zhang, 2023)
 - ENSO predictions for 18 months
 - Transformers surpass CNNs
- Parameterization for Vertical Mixing – FCNN (Zhu et al., 2022)
 - Physically constrained and embedded in a climate model



Timeline Review

- ENSO prediction - CNN + Transfer Learning (2019) *AlexNet 2012*
- Pangu-Weather, GeoFormer, XiHe – Transformers (2023-2024) *– 7yrs*
AAVN 2017 - 6yrs
- GraphCast – GNN (2023) *GraphSage 2017 – 6yrs*
- FourCastNet – AFNO (2023) *Li et al. 2020 – 3yrs*

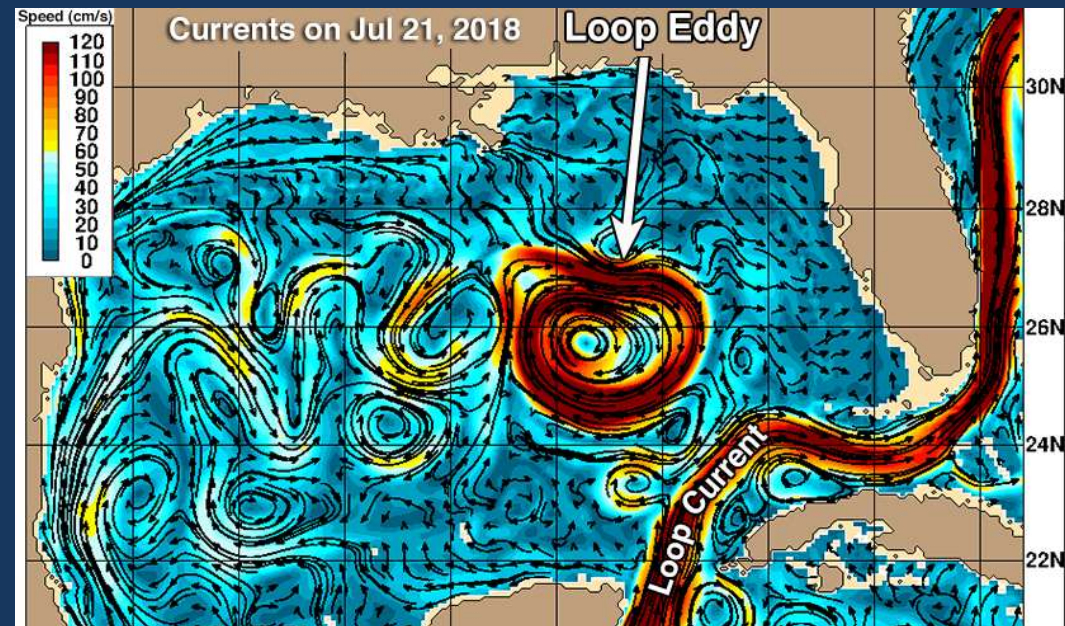
Also...

- Diffusion: DDPMs (2020) vs DIFFDS for SST (2024) – *4yrs*
- Foundation Modeling: ViT, GPT4 (2022) vs AlphaOcean/IBM/UH - *3yrs*

Difficulties in Marine Science

Massive amounts of data, but
“event sparse”

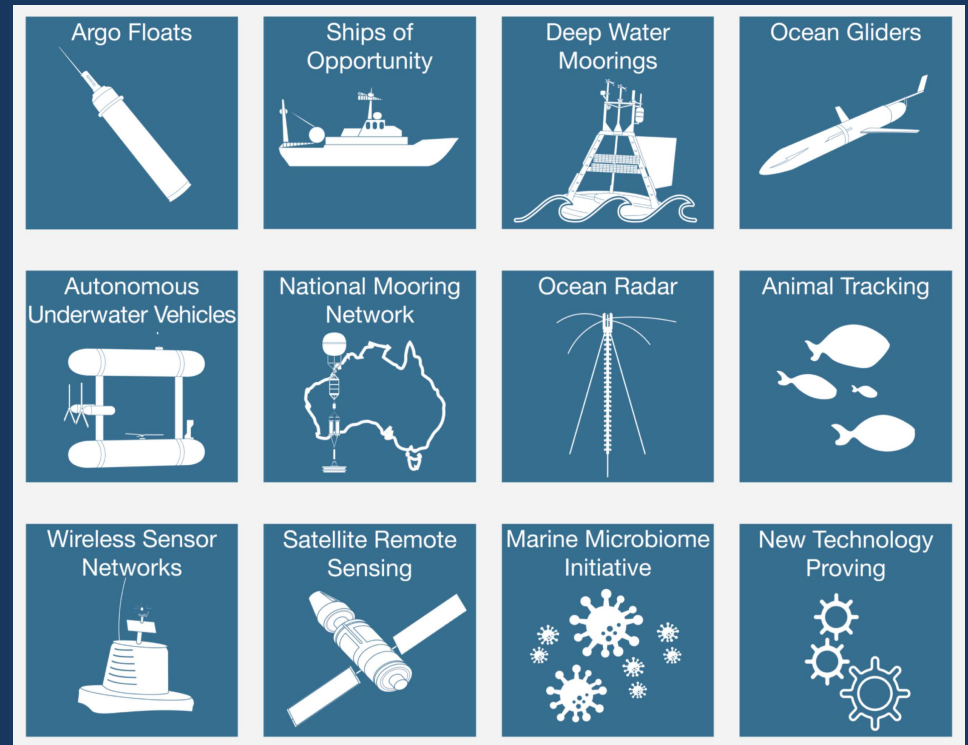
- Satellites, model output, ship tracks, buoys, etc. Petabytes
- *Interesting* signals and events are rare
 - Eddy events
 - Heatwaves
 - Harmful Algal Blooms



Difficulties in Marine Science

Data is heterogeneous and asynchronous

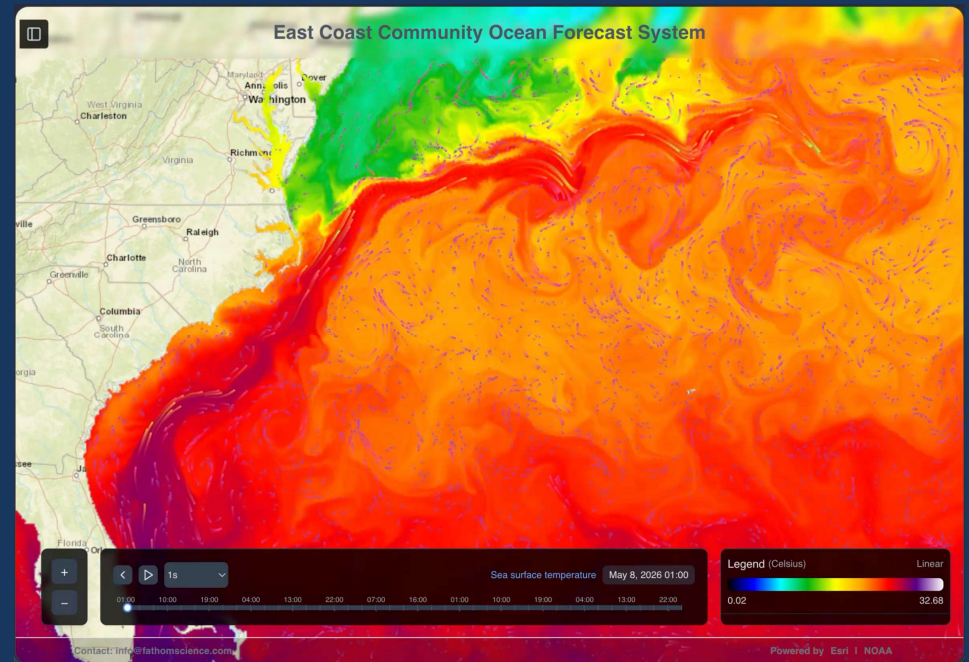
- Satellites are gridded, 2D, may be in “swaths”
- Ship tracks are random shapes
- Buoys are single points
- Some instruments take vertical, 1D profiles
- All measurements on independent sampling period



Difficulties in Marine Science

Multi-scale features in almost all datasets

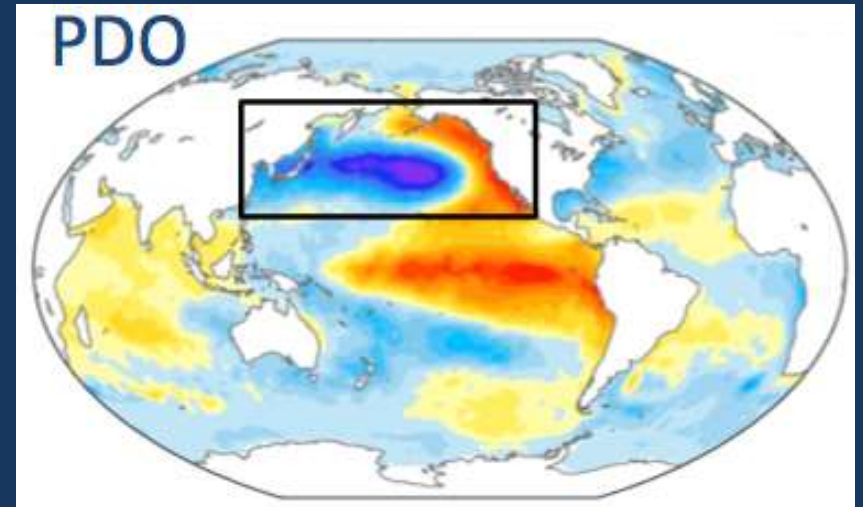
- Meter-scale turbulence to 1000km-scale gyres
- All features coupled
- CNNs and ViTs struggle at such a range of scale



Difficulties in Marine Science

Slow evolution and sparse time series

- Compared to atmosphere: greater density, friction, specific heat
- Time scales of events are much much longer – can be decades in length
- Instruments for measurement have only existed for a few decades so sample size is small

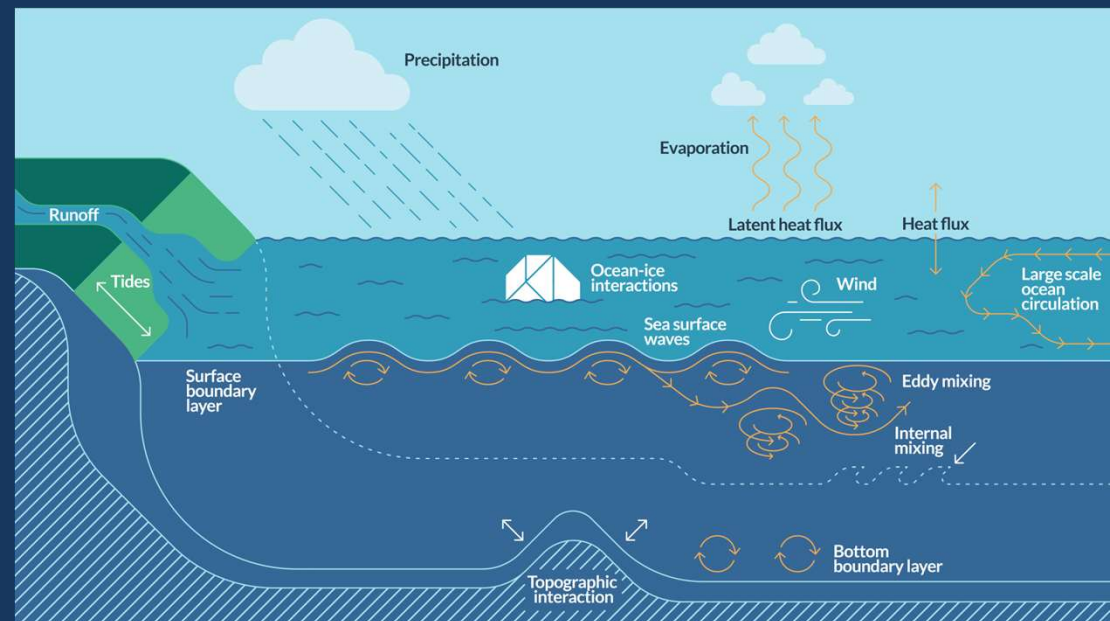


Difficulties in Marine Science

Physics is everything

- Results can look similar, but may hold physically inconsistent fields
- Low RMSE does not mean mass, momentum, vorticity, etc. are conserved
 - Averaged fields in general are prone to miss the physics

If it's not physical, it's wrong



What do we do about it?

Focus on the physics

- If you get the physics right, the rest follows
- Physics Informed Neural Networks
- Numerical (physical) models + AI parameterization models
- For forecasts, ensemble modeling (probabilistic; error estimating)
- Data Assimilation: Combine data into better data!

Progress being made

NOAA Releases 10-Year Strategy for Data Assimilation

📅 February 26, 2025

[Submitted on 20 Mar 2026]

Data-driven ensemble prediction of the global ocean

Qiusheng Huang, Xiaohui Zhong, Anboyu Guo, Ziyi Peng, Lei Chen, Hao Li

[Submitted on 26 Oct 2025]

A Framework for Hybrid Physics–AI Coupled Ocean Models

Laure Zanna, William Gregory, Pavel Perezhogin, Aakash Sane, Cheng Zhang, Alistair Adcroft, Mitch Bushuk, Carlos Fernandez-Granda, Brandon Reichl, Dhruv Balwada, Julius Busecke, William Chapman, Alex Connolly, Danni Du, Kelsey Everard, Fabrizio Falasca, Renaud Falga, David Kamm, Etienne Meunier, Qi Liu, Antoine Nasser, Matthew Pudig, Andrew Shao, Julia L. Simpson, Linus Vogt, Jiarong Wu

Front. Mar. Sci., 09 September 2025

Sec. Physical Oceanography

Volume 12 - 2025 | <https://doi.org/10.3389/fmars.2025.1547995>

Physics-informed neural networks to reconstruct surface velocity field from drifter data

Technical Bottlenecks

Compute Access

- Right?
- GPUs are expensive
- Individual training samples still large enough to be “slow” on GPUs

Technical Bottlenecks

Storage and Access

- Expensive to store terabytes+ of data locally
- Storing large datasets online typically means slow or rate-limited access
- Possible solutions, such as S3, require filetypes (HDF5, NetCDF4, Zarr) that aren't natively compatible with ML libraries



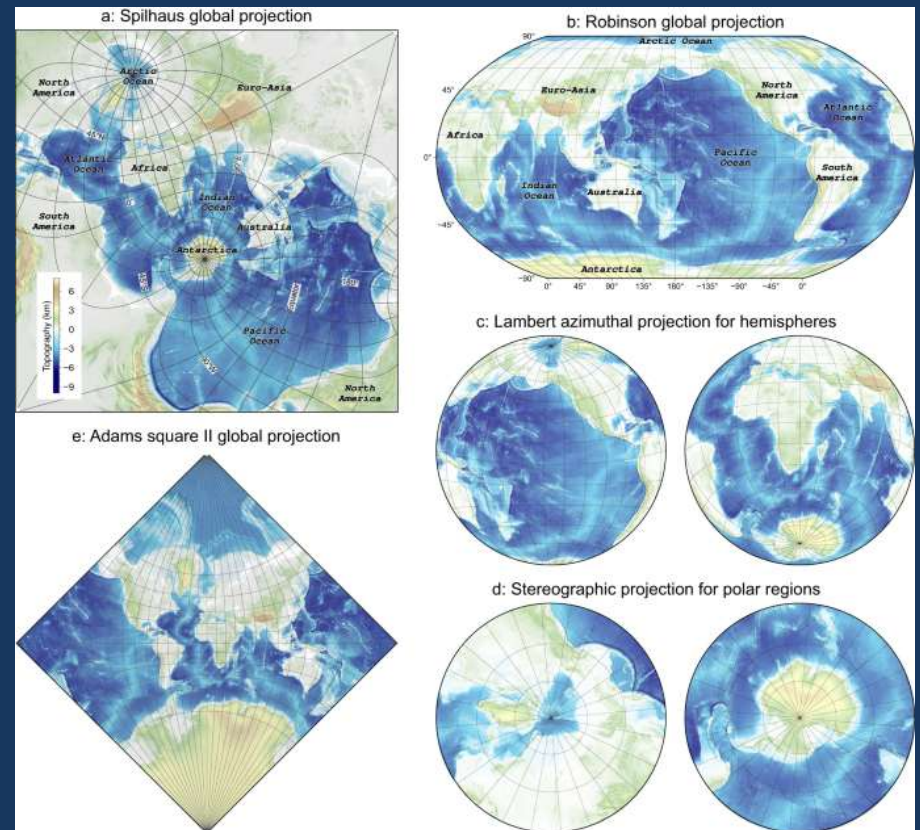
Amazon
S3



Technical Bottlenecks

Data preprocessing can be complicated

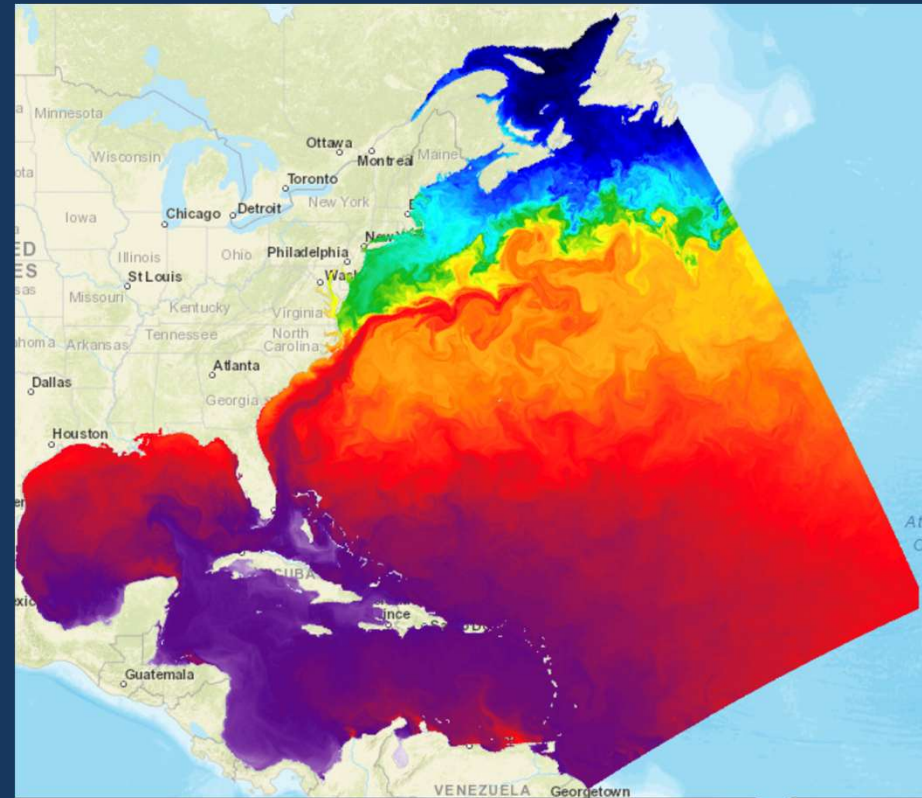
- Datasets may require regridding, interpolation, projections
- Observational datasets may have large gaps in
 - Especially true in satellite data



Technical Bottlenecks

Simulated (numerical model) data is expensive and slow

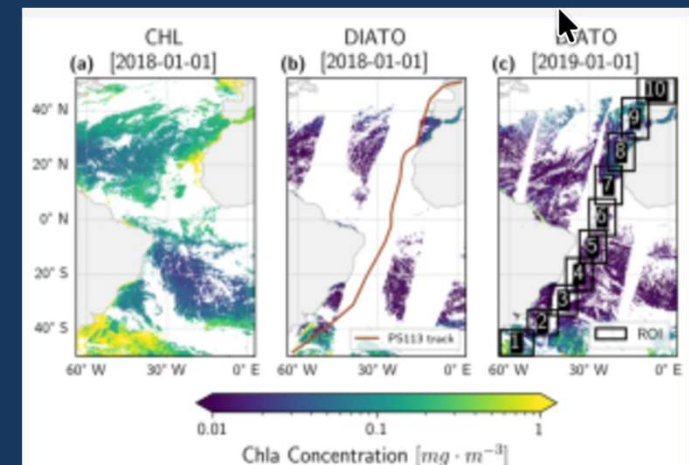
- Can be used as “truth” during training
- Complete and not spotty like observations may be
- So expensive and so slow that we are turning to AI



Technical Bottlenecks

True validation requires observations

- Models of any kind aren't considered "ground truth"
- Observations are the ultimate seal of approval
- Spotty, sparse data doesn't typically make for good data to train against
 - If used, entire model gets updated based on the error of a "few" points



Technical Bottlenecks

A lot of solutions require an Oceanographer in the loop

- Selecting loss functions, choosing numerical models, handling geophysical data formats
- Some events are subtle in data without a trained eye

NAVIER-STOKES EQUATION

$$\rho g_x - \frac{\partial p}{\partial x} + \mu \left(\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} + \frac{\partial^2 u}{\partial z^2} \right) = \rho \frac{Du}{Dt}$$

$$\rho g_y - \frac{\partial p}{\partial y} + \mu \left(\frac{\partial^2 v}{\partial x^2} + \frac{\partial^2 v}{\partial y^2} + \frac{\partial^2 v}{\partial z^2} \right) = \rho \frac{Dv}{Dt}$$

$$\rho g_z - \frac{\partial p}{\partial z} + \mu \left(\frac{\partial^2 w}{\partial x^2} + \frac{\partial^2 w}{\partial y^2} + \frac{\partial^2 w}{\partial z^2} \right) = \rho \frac{Dw}{Dt}$$

Navier-Stokes Equation

$$\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \vec{U}) = 0$$

Continuity

What do we do about it?

The problems at the intersection of AL and marine science require collaboration between Computer/AI scientists and Oceanographers

Computer Scientists

- Architectural design
 - Familiarity with different categories of ML models
- Scalable training
 - Efficient code for large data samples that scale to even larger datasets
- Familiarity with cloud-optimized storage and data access

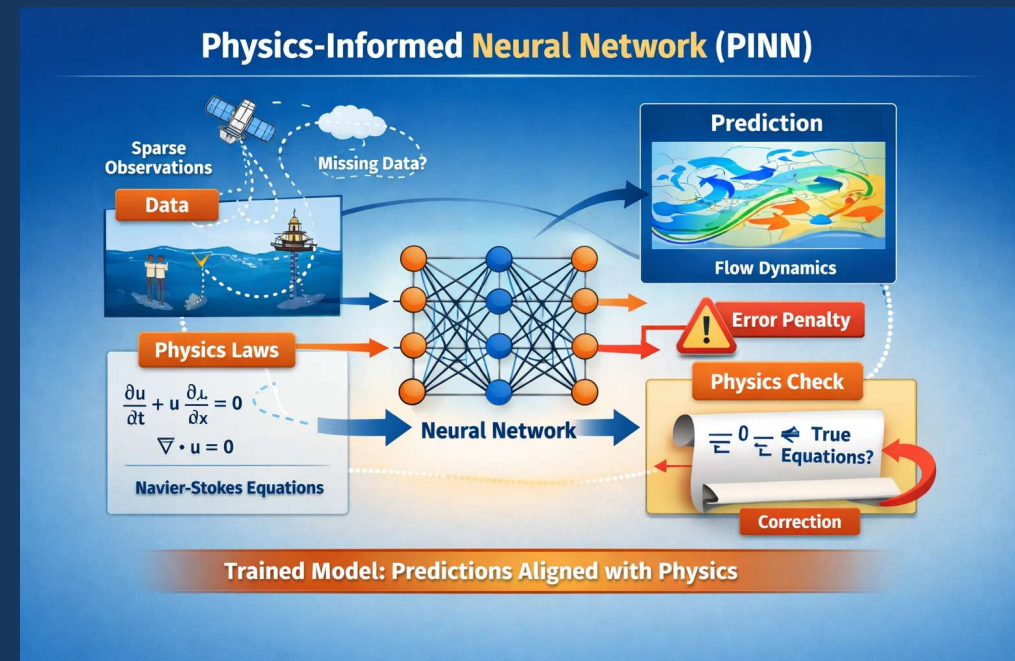
Marine Scientists

- Domain expertise
 - How to manipulate different variables
- Data source access
 - Data science side of large data
- Familiarity with literature
 - Use trends in literature to help decide on new features
- A healthy dose of skepticism



Collaborations

- Creating/standardizing ML-ready datasets
- Physics-aware models
- Benchmarking strategies and pipelines

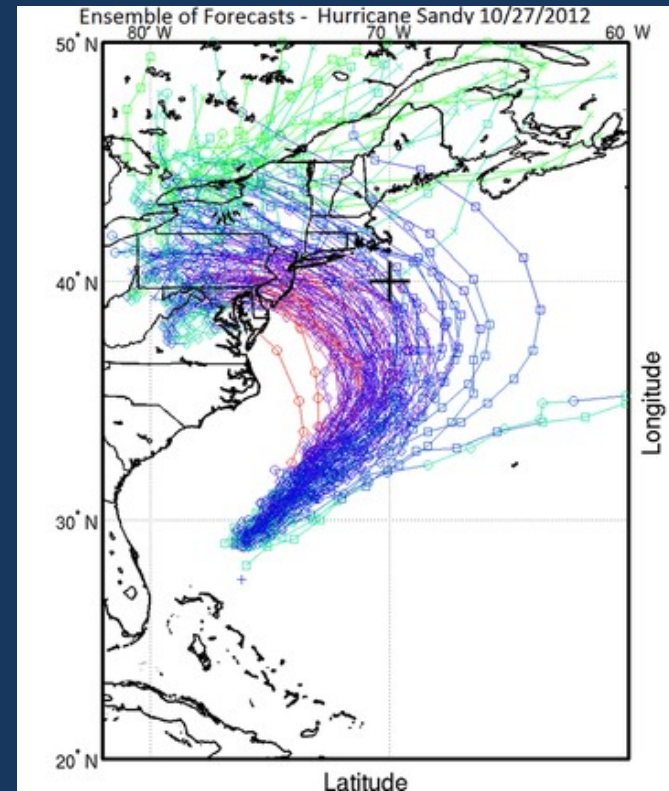


On the way...

- IceNet – Unet ensemble (Andersson et al., 2021)
 - Monthly sea ice concentrations
 - First model with uncertainty awareness
- Parameterization for Vertical Mixing – FCNN (Zhu et al., 2022)
 - Physically constrained and embedded in a climate model
- Data Assimilation – Diffusion (Martin et al. 2025)
 - Gap filling satellite data

On the way...

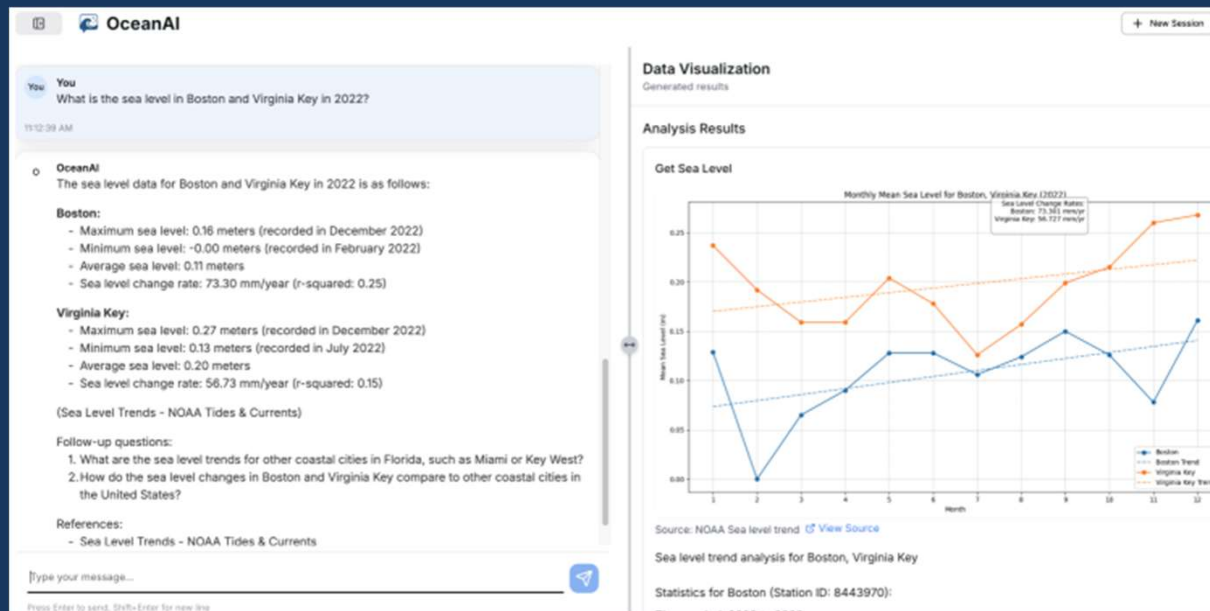
- Ensemble modeling
 - Suppresses chaos, probabilistic
 - Provided uncertainty estimates
- Parameterization
 - Replace long running portions of numerical models with AI models
- Improving data quality to increase training samples of all models



Projects at Fathom Science

Ocean AI – LLM catered towards answering questions about oceanography

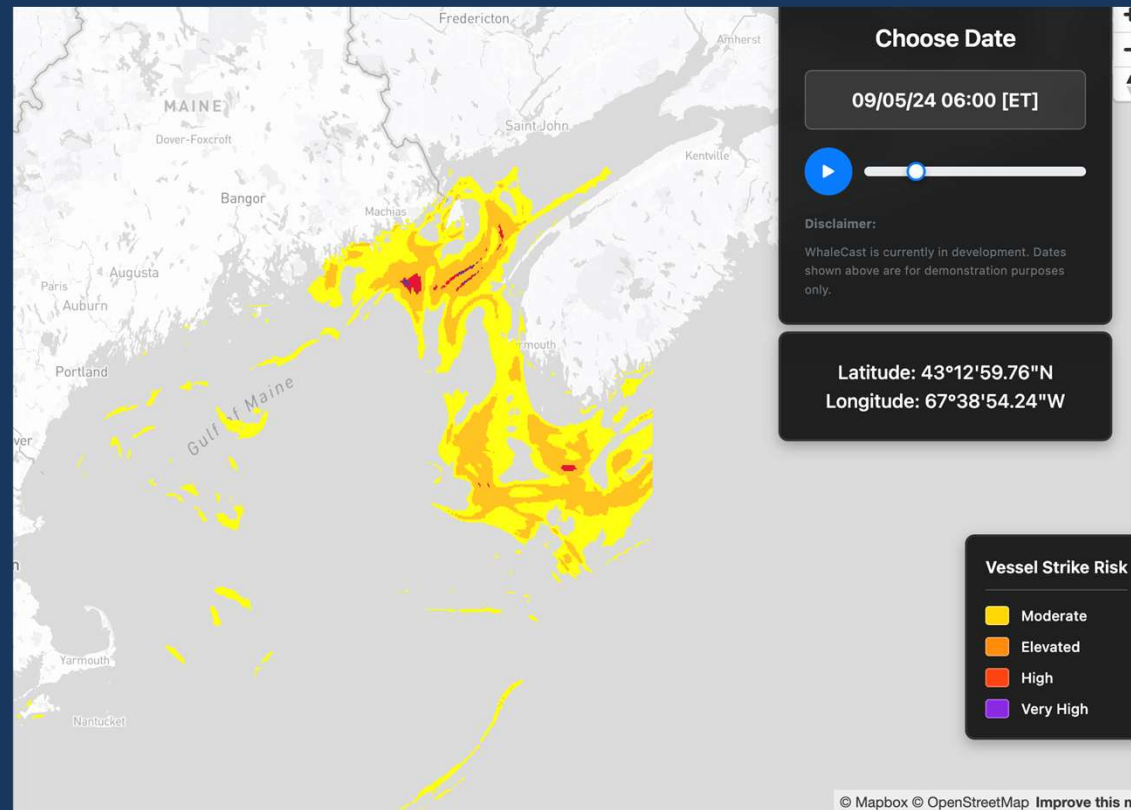
- w/ NCSU



Projects at Fathom Science

WhaleCast

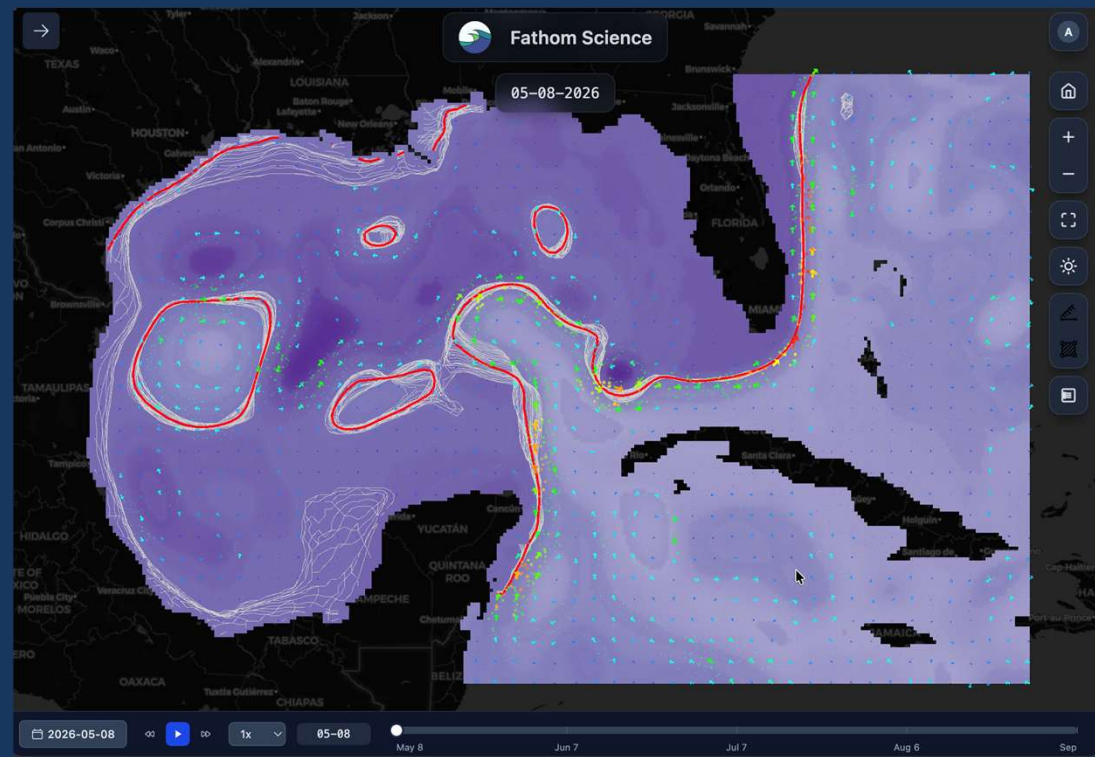
- Predicting likelihood of whale sightings to help avoid vessel strikes
- Data-driven: documented sightings vs ocean reanalysis



Projects at Fathom Science

OceanNet

- My thesis, iterated on
- Multiple variables, longer prediction horizon
- Depth
- Ensemble predictions

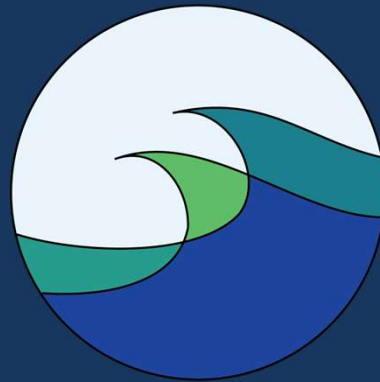


Conclusion

- Marine Science has advanced its AI capabilities at a steady pace over the past decade
- Despite these advances, there remains many challenges regarding data, training pipelines, and theory
- As AI/ML research is advancing rapidly, it is imperative for computer/AI scientist to work with Oceanographers to overcome computing and data difficulties
- While computer scientists improve architectural designs, scalability, and training pipelines, oceanographers can work to derive better methods for incorporating physics into AI models and creating methods for models to compliment numerical models

Thank You

Website



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